

Continuum Mechanics HW3

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Technion

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1 Jet Incident onto a Plane

1.1 force on surface

The incident velocity is, expressed as a vector

$$\mathbf{v}_{in} = U (-\cos \theta, -\sin \theta) \quad (1)$$

and the outgoing velocities are

$$\mathbf{U}_1 = (-U_1, 0), \quad \mathbf{U}_2 = (U_2, 0) \quad (2)$$

from conservation of mass

$$Uh = U_1 h_1 + U_2 h_2 \quad (3)$$

so

$$h = h_1 + h_2 \quad (4)$$

using Bernoulli's theorem since the flow is steady and ideal,

$$p_0 + \frac{1}{2}\rho U^2 = p_0 + \frac{1}{2}\rho U_1^2 = p_0 + \frac{1}{2}\rho U_2^2 \quad (5)$$

so

$$U = U_1 = U_2 \quad (6)$$

the momentum flux going into the system is

$$\dot{\mathbf{P}}_{in} = \rho h U^2 (-\cos \theta, -\sin \theta) \quad (7)$$

and the momentum flux leaving the system is

$$\dot{\mathbf{P}}_{out} = \rho U^2 (h_2 - h_1, 0) \quad (8)$$

so the force the plate applies on the liquid is

$$\mathbf{F} = \dot{\mathbf{P}}_{out} - \dot{\mathbf{P}}_{in} = \rho U^2 (h_2 - h_1 + h \cos \theta, h \sin \theta) \quad (9)$$

so

$$\mathbf{f} = \rho U^2 ((h_1 - h_2 - h \cos \theta) \hat{\mathbf{x}} - h \sin \theta \hat{\mathbf{y}}) \quad (10)$$

Since the jet is of an ideal fluid, the only stress it can transmit to the plate is from pressure, which always acts along the normal. so,

$$\mathbf{f} = -\rho U^2 - h \sin \theta \hat{\mathbf{y}} \quad (11)$$

1.2 thickness as a function

we already know

$$h = h_1 + h_2 \quad (12)$$

and we can add

$$h_1 - h_2 = h \cos \theta \quad (13)$$

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from the parallel part of the force being zero. We can now insert the above relations into each other and get

$$h - h_2 - h_2 = h \cos \theta \implies h_2 = \frac{h}{2} (1 - \cos \theta) \quad (14)$$

$$h_1 - h + h_1 = h \cos \theta \implies h_1 = \frac{h}{2} (1 + \cos \theta) \quad (15)$$

2 Boundary Layer on a Surface Attached to a Sink

2.1 Boundary conditions

The boundary conditions are non-penetration, no slip, and the condition to match the outer flow:

$$v_y(x > 0, y = 0) = 0 \quad (16)$$

$$v_x(x > 0, y = 0) = 0 \quad (17)$$

$$v_x(x, y) \rightarrow U(x) = \frac{m}{x} \text{ as } y/\delta(x) \rightarrow \infty \quad (18)$$

2.2 thickness

The second principle is that inside the layer the inertial term and the viscous terms are of the same order. We now look at the approximations

$$v_x \sim U, \quad \partial_x \sim 1/x, \quad \partial_y \sim 1/\delta \quad (19)$$

which give

$$v_x \frac{\partial v_x}{\partial x} \sim \frac{U^2}{x}, \quad \nu \frac{\partial^2 v_x}{\partial y^2} \sim \frac{\nu U}{\delta^2} \quad (20)$$

balancing them

$$\frac{U^2}{x} \sim \frac{\nu U}{\delta^2} \Rightarrow \delta \sim \sqrt{\frac{\nu x}{|U(x)|}} = \sqrt{\frac{\nu x}{|m|/x}} \Rightarrow \delta(x) = x \sqrt{\frac{\nu}{|m|}} \propto x \quad (21)$$

2.3 showing equation

The stream function from class is

$$\Psi(x, y) = -U(x)\delta(x)f(\eta) \quad (22)$$

but

$$U(x)\delta(x) = \frac{m}{x} x \sqrt{\frac{\nu}{|m|}} = \sqrt{\nu|m|} \quad (23)$$

which is constant, meaning Ψ is a function of η alone.